



Important: Text generated by AI and may contain incorrect information, always check the content.
To understand the main characteristics of the dataset, we need to analyze the structure, data types, and summary statistics of the given DataFrame.

```
```python
Import necessary libraries
import pandas as pd

First, let's check the shape and data types of the DataFrame
display(df.head())
display(df.shape)
display(df.dtypes)

Check for missing values
display('Display of missing values:')
display(df.isnull().sum())

Summary statistics for numerical columns
display('Summary statistics for numerical columns:')
display(df.describe())
```
```

| | Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | ... | AnyHealthcare | NoDocbcCost | GenHlth | MentHlth | PhysHlth | DiffWalk | Sex | Age | Education | Income |
|---|--------------|--------|----------|-----------|------|--------|--------|----------------------|--------------|--------|-----|---------------|-------------|---------|----------|----------|----------|-----|------|-----------|--------|
| 0 | 0.0 | 1.0 | 1.0 | 1.0 | 40.0 | 1.0 | 0.0 | 0.0 | 0.0 | 0.0 | ... | 1.0 | 0.0 | 5.0 | 18.0 | 15.0 | 1.0 | 0.0 | 9.0 | 4.0 | 3.0 |
| 1 | 0.0 | 0.0 | 0.0 | 0.0 | 25.0 | 1.0 | 0.0 | 0.0 | 1.0 | 0.0 | ... | 0.0 | 1.0 | 3.0 | 0.0 | 0.0 | 0.0 | 0.0 | 7.0 | 6.0 | 1.0 |
| 2 | 0.0 | 1.0 | 1.0 | 1.0 | 28.0 | 0.0 | 0.0 | 0.0 | 0.0 | 1.0 | ... | 1.0 | 1.0 | 5.0 | 30.0 | 30.0 | 1.0 | 0.0 | 9.0 | 4.0 | 8.0 |
| 3 | 0.0 | 1.0 | 0.0 | 1.0 | 27.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | ... | 1.0 | 0.0 | 2.0 | 0.0 | 0.0 | 0.0 | 0.0 | 11.0 | 3.0 | 6.0 |
| 4 | 0.0 | 1.0 | 1.0 | 1.0 | 24.0 | 0.0 | 0.0 | 0.0 | 1.0 | 1.0 | ... | 1.0 | 0.0 | 2.0 | 3.0 | 0.0 | 0.0 | 0.0 | 11.0 | 5.0 | 4.0 |



5 rows × 22 columns
(253680, 22)

| | |
|----------------------|---------|
| | 0 |
| Diabetes_012 | float64 |
| HighBP | float64 |
| HighChol | float64 |
| CholCheck | float64 |
| BMI | float64 |
| Smoker | float64 |
| Stroke | float64 |
| HeartDiseaseorAttack | float64 |
| PhysActivity | float64 |
| Fruits | float64 |
| Veggies | float64 |
| HvyAlcoholConsump | float64 |
| AnyHealthcare | float64 |





| 0 | |
|-------------|---------|
| NoDocbcCost | float64 |
| GenHlth | float64 |
| MentHlth | float64 |
| PhysHlth | float64 |
| DiffWalk | float64 |
| Sex | float64 |
| Age | float64 |
| Education | float64 |
| Income | float64 |

dtype: object
'Display of missing values: '

| 0 | |
|----------------------|---|
| Diabetes_012 | 0 |
| HighBP | 0 |
| HighChol | 0 |
| CholCheck | 0 |
| BMI | 0 |
| Smoker | 0 |
| Stroke | 0 |
| HeartDiseaseorAttack | 0 |
| PhysActivity | 0 |
| Fruits | 0 |
| Veggies | 0 |
| HvyAlcoholConsump | 0 |
| AnyHealthcare | 0 |
| NoDocbcCost | 0 |
| GenHlth | 0 |
| MentHlth | 0 |
| PhysHlth | 0 |
| DiffWalk | 0 |
| Sex | 0 |





| | 0 |
|-----------|---|
| Age | 0 |
| Education | 0 |
| Income | 0 |

dtype: int64
'Summary statistics for numerical columns:'

| | Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | ... | AnyHealthcare | |
|-------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|----------------------|---------------|---------------|-----|---------------|---------------|
| count | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | 253680.000000 | ... | 253680.000000 | 253680.000000 |
| mean | 0.296921 | 0.429001 | 0.424121 | 0.962670 | 28.382364 | 0.443169 | 0.040571 | 0.094186 | 0.756544 | 0.634256 | ... | 0.951053 | |
| std | 0.698160 | 0.494934 | 0.494210 | 0.189571 | 6.608694 | 0.496761 | 0.197294 | 0.292087 | 0.429169 | 0.481639 | ... | 0.215759 | |
| min | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 12.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | 0.000000 | ... | 0.000000 | |
| 25% | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 24.000000 | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 0.000000 | ... | 1.000000 | |
| 50% | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 27.000000 | 0.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |
| 75% | 0.000000 | 1.000000 | 1.000000 | 1.000000 | 31.000000 | 1.000000 | 0.000000 | 0.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |
| max | 2.000000 | 1.000000 | 1.000000 | 1.000000 | 98.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | 1.000000 | ... | 1.000000 | |

8 rows x 22 columns
By examining the output of these operations, we can determine the number of rows and columns, data types of each column, presence or absence of missing values, and summary statistics such as mean, standard deviation, etc.
To further analyze the characteristics of the dataset, we should also consider the distribution of the data.

```
```python
Distribution of numerical columns
display('Distribution of numerical columns:')
for col in df.columns:
 display(f'Distribution of {col}:')
 display(df[col].value_counts().head())
```
```

'Distribution of numerical columns:'
'Distribution of Diabetes_012:'

| | Diabetes_012 |
|-----|--------------|
| 0.0 | 213703 |
| 2.0 | 35346 |
| 1.0 | 4631 |

dtype: int64
'Distribution of HighBP:'

| | HighBP |
|-----|--------|
| 0.0 | 144851 |
| 1.0 | 108829 |

dtype: int64
'Distribution of HighChol:'





| HighChol | |
|----------|--------|
| 0.0 | 146089 |
| 1.0 | 107591 |

dtype: int64
'Distribution of CholCheck: '

| CholCheck | |
|-----------|--------|
| 1.0 | 244210 |
| 0.0 | 9470 |

dtype: int64
'Distribution of BMI: '

| BMI | |
|------|-------|
| 27.0 | 24606 |
| 26.0 | 20562 |
| 24.0 | 19550 |
| 25.0 | 17146 |
| 28.0 | 16545 |

dtype: int64
'Distribution of Smoker: '

| Smoker | |
|--------|--------|
| 0.0 | 141257 |
| 1.0 | 112423 |

dtype: int64
'Distribution of Stroke: '

| Stroke | |
|--------|--------|
| 0.0 | 243388 |
| 1.0 | 10292 |

dtype: int64
'Distribution of HeartDiseaseorAttack: '

| HeartDiseaseorAttack | |
|----------------------|--------|
| 0.0 | 229787 |
| 1.0 | 23893 |

dtype: int64
'Distribution of PhysActivity: '

| PhysActivity | |
|--------------|--------|
| 1.0 | 191920 |
| 0.0 | 61760 |





dtype: int64
'Distribution of Fruits: '

| Fruits | |
|--------|--------|
| 1.0 | 160898 |
| 0.0 | 92782 |

dtype: int64
'Distribution of Veggies: '

| Veggies | |
|---------|--------|
| 1.0 | 205841 |
| 0.0 | 47839 |

dtype: int64
'Distribution of HvyAlcoholConsump: '

| HvyAlcoholConsump | |
|-------------------|--------|
| 0.0 | 239424 |
| 1.0 | 14256 |

dtype: int64
'Distribution of AnyHealthcare: '

| AnyHealthcare | |
|---------------|--------|
| 1.0 | 241263 |
| 0.0 | 12417 |

dtype: int64
'Distribution of NoDocbcCost: '

| NoDocbcCost | |
|-------------|--------|
| 0.0 | 232326 |
| 1.0 | 21354 |

dtype: int64
'Distribution of GenHlth: '

| GenHlth | |
|---------|-------|
| 2.0 | 89084 |
| 3.0 | 75646 |
| 1.0 | 45299 |
| 4.0 | 31570 |
| 5.0 | 12081 |

dtype: int64
'Distribution of MentHlth: '





| MentHlth | |
|----------|--------|
| 0.0 | 175680 |
| 2.0 | 13054 |
| 30.0 | 12088 |
| 5.0 | 9030 |
| 1.0 | 8538 |

dtype: int64
'Distribution of PhysHlth:'

| PhysHlth | |
|----------|--------|
| 0.0 | 160052 |
| 30.0 | 19400 |
| 2.0 | 14764 |
| 1.0 | 11388 |
| 3.0 | 8495 |

dtype: int64
'Distribution of DiffWalk:'

| DiffWalk | |
|----------|--------|
| 0.0 | 211005 |
| 1.0 | 42675 |

dtype: int64
'Distribution of Sex:'

| Sex | |
|-----|--------|
| 0.0 | 141974 |
| 1.0 | 111706 |

dtype: int64
'Distribution of Age:'

| Age | |
|------|-------|
| 9.0 | 33244 |
| 10.0 | 32194 |
| 8.0 | 30832 |
| 7.0 | 26314 |
| 11.0 | 23533 |

dtype: int64
'Distribution of Education:'





| Education | |
|-----------|--------|
| 6.0 | 107325 |
| 5.0 | 69910 |
| 4.0 | 62750 |
| 3.0 | 9478 |
| 2.0 | 4043 |

dtype: int64
'Distribution of Income: '

| Income | |
|--------|-------|
| 8.0 | 90385 |
| 7.0 | 43219 |
| 6.0 | 36470 |
| 5.0 | 25883 |
| 4.0 | 20135 |

dtype: int64
This will provide insights into the distribution of values in each column, helping to identify any patterns or anomalies.
Additionally, we can check the correlation between different numerical columns to understand their relationships.

```
```python
Correlation between numerical columns
display('Correlation between numerical columns:')
display(df.corr().head())
```
```

'Correlation between numerical columns: '

| | Diabetes_012 | HighBP | HighChol | CholCheck | BMI | Smoker | Stroke | HeartDiseaseorAttack | PhysActivity | Fruits | ... | AnyHealthcare | NoDocbcCost | GenHlth | MentHlth | PhysHlth |
|--------------|--------------|----------|----------|-----------|----------|-----------|----------|----------------------|--------------|-----------|-----|---------------|-------------|----------|-----------|----------|
| Diabetes_012 | 1.000000 | 0.271596 | 0.209085 | 0.067546 | 0.224379 | 0.062914 | 0.107179 | 0.180272 | -0.121947 | -0.042192 | ... | 0.015410 | 0.035436 | 0.302587 | 0.073507 | 0.176287 |
| HighBP | 0.271596 | 1.000000 | 0.298199 | 0.098508 | 0.213748 | 0.096991 | 0.129575 | 0.209361 | -0.125267 | -0.040555 | ... | 0.038425 | 0.017358 | 0.300530 | 0.056456 | 0.161212 |
| HighChol | 0.209085 | 0.298199 | 1.000000 | 0.085642 | 0.106722 | 0.091299 | 0.092620 | 0.180765 | -0.078046 | -0.040859 | ... | 0.042230 | 0.013310 | 0.208426 | 0.062069 | 0.121751 |
| CholCheck | 0.067546 | 0.098508 | 0.085642 | 1.000000 | 0.034495 | -0.009929 | 0.024158 | 0.044206 | 0.004190 | 0.023849 | ... | 0.117626 | -0.058255 | 0.046589 | -0.008366 | 0.031775 |
| BMI | 0.224379 | 0.213748 | 0.106722 | 0.034495 | 1.000000 | 0.013804 | 0.020153 | 0.052904 | -0.147294 | -0.087518 | ... | -0.018471 | 0.058206 | 0.239185 | 0.085310 | 0.121141 |

5 rows × 22 columns

